Problem Statement and Goals
Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
September 28, 2026	All	Initial Revision for Deliverable 1

1 Problem Statement

1.1 Problem

Around 440,000 Canadians deal with some kind of speech, language, or communication issue [1]; this includes 7-10% of preschool children in Canada [2]. A common speech impediment in children is substitution, which is when an expected phoneme is replaced by another (e.g. a child may pronounce "Rabbit" as "Wabbit") [3]. For detection of said issues, speech language pathologists (SLPs) and researchers have used various tools for analyzing speech and helping individuals improve. Current phonetic detection tools for researchers and SLPs are proprietary, not compliant with the Personal Health Information Protection Act (PHIPA), and/or poor in performance (e.g. speed and accuracy) [4]. This means that available tools are insufficient for research and for treating substitution speech impediments in children. There should be a substitution treatment tool for child SLPs that is open-source, PHIPA-compliant, performant, and expandable to other professions (e.g. accent coaches) and other speech differences beyond execution.

1.2 Inputs and Outputs

1.2.1 Inputs

Patient side

- Patient's speech patterns: When a patient is in a session with specialist, they will be prompted to speak into a microphone that will send the data to the system for phonetic processing.
- Patient's login credentials: If a patient wants to view their progress outside of a session or do take-home exercises, they will need to access their data by logging into the system with their credentials.

Specialist side

- Specialist's login credentials: The specialist will need to be able to log in to the administration side of the system to access lessons and patient data. They will do so by logging in with their credentials.
- Specialist's Selection of Profession: The specialist selects what their profession is, as the system is uniquely tuned to a specialist's needs based on their selected profession (SLP, Accent Coach, Researcher, etc.).
- Patient Selection: The specialist will need to use the system to access any given patient's data to identify problems and track progress.
- Model Selection: The specialist will select the type of speech impediment that they want to work on with a patient to improve specific phonemes.

Both Sides

- Select unavailable phonemes: Some patients will be physically unable to use certain phonemes based on biological or neurological limitations. Thus, the specialist for in-person sessions and the patient for homework or out-of-clinic sessions will select which phonemes the patient cannot produce.

1.2.2 Outputs

Both Sides

- Phonemic speech analysis: During a session, after each exercise and the phonemic analysis is complete, the system will display the phonemes it perceived from the patient and compare it against the phonemes it expected to hear from the exercise. This information will be available to both the patient and specialist to track progress.

- Patient data and progress tracking: The system will be able to display the patient's progress on improving their pronunciation of phonemes. It will show all the areas the patient is excelling at and which areas they need to improve on. This data will be a simplified version of the overall metrics of success used by the system so patients can easily understand how they are progressing. This also helps to prevent hackers from harvesting all of the sensitive data from a compromised account.

- Suggested lessons for improvement: While the patient is in-session with the specialist or while they are at home, the system can recommend lessons for them to do to improve their phoneme pronunciation. The choice to use lessons recommended by the system is strictly up to the specialist and/or patient.

Specialist Side

- Detailed metrics and data: The system will output the full range of data for specialists so that they can engage in a deeper analysis of a given patient's progress.

1.3 Stakeholders

1.3.1 Primary Stakeholders

Speech Language Pathologists - Speech Language Pathologists are the main user group for this application. Their job revolves around correcting speech disorders, so this tool would serve as their assistant during everyday activities. It may be used to assess speech patterns, and to discover what phonemes are being misspoken. They will be able to use the tool's feedback to speed up and adjust treatment as necessary.

Patients - Patients (primarily children) would interact with this application very often. Their improvement is directly tied to how accurately the application assesses their speaking when providing feedback to a clinician. Due to their young age, it is assumed that they have little digital literacy and education. This is a key consideration when designing an interface for them to complete homework, or when giving them feedback.

1.3.2 Secondary Stakeholders

Parents and Guardians - Patient Guardians are not the main user group for this application but hold a large interest in how it performs. For them to trust that it is giving their dependent person good treatment, it needs to be accurate. It must also keep their personal data safe. Besides indirect concerns, they may be occasional users when tracking their child's progress, or when assisting them with homework, making them an important consideration.

1.3.3 Tertiary Stakeholders

Academia (Linguistics Students, Cognitive Science Researchers) - There are several uses for this application in academia. This tool may be leveraged to extract and analyze pediatric speech to uncover hidden patterns during research. It may also be used as a tool for aspiring SLPs to reference when learning speech language pathology.

Governments - In Ontario, PHIPA protects patient confidentiality. Therefore, local governments would be concerned with how the application maintains patient data, to properly protect their privacy.

1.4 Environment

Hardware:

- PC (desktop or laptop, Specs to be determined)
- Microphone (built-in or external)

Software/Firmware:

- Web browser (Google Chrome or Edge)
- Windows 10 and 11 Operating System
- Any required drivers for microphone
- Supports HTTPS connections

2 Goals

The overall objective of this project is to provide assistance to pediatric speech-language pathologists during their patient sessions. This will be a software tool that can act as an in-session assistant to pediatric speech-language pathologists (SLPs) during speech therapy sessions, helping them identify speech challenges quickly and tailor therapy more effectively. The system will be strictly assistive, augmenting the clinician's expertise but not replacing their professional judgement.

Listed below are the overall goals for this project:

- Develop a web-based software tool that functions as an in-session assistant of pediatric SLPs.
- Analyze real-time speech of a patient at the phonetic level to identify and flag speech substitution speech patterns, a common pediatric speech challenge where an expected phoneme is replaced by another.

- Include an interactive interface for SLPs, allowing them to record and analyze speech during session, including insights and data visualizations on produced sounds and identified speech pattern challenges.
- Include features and activities within the interface that the SLP can lead a patient through to help them improve on speech challenges during a session.

To achieve these goals there are two primary components:

1. **Speech Analysis Model:** The core technical component of the system. It will be responsible for processing audio input and extracting phonetic information from speech in real-time. The goals of this component are:
    - Accept real-time audio input of a patient's speech during a therapy session.
    - Transcribe and classify the speech of a patient at the phonetic level.
    - Identify substitution speech patterns.
    - Operate with sufficient speed and accuracy to be useful during a live speech therapy session.
2. **SLP User Interface:** The "assistant" component of the system, interacted with by the SLP. It will be responsible for visually representing analysis results and supporting the SLP's workflow during the session. The goals of this component are:
    - Provide an interactive dashboard for the SLP to record a patient's speech in real-time.
    - Display data visualisations of the sounds produced, and flagged speech patterns.
    - Present session data in a way that supports the SLP's clinical process.
    - Offer guided activities and exercises that the SLP can provide to the patient during sessions to target a patient's specific speech challenges.
    - Display session tracking and patient progress data

3 Stretch Goals

The following are goals for possible expansion of the project after the main goals are achieved:

- **Patient-Facing/Homework Interface:** An additional interface for patients, allowing then to complete at home practice activities to improve their speech.
    - Account for different types of users (SLP & patients).

- **Mobile Application:** An Android/iOS application for patients to practice their speech and complete homework activities on a mobile device.
- **Broader Speech Pattern Detection:** The detection of additional types of speech pattern challenges beyond just substitution.
  - Cluster reduction and omisison patterns.
  - Distortion patterns.
- **Additional Interfaces:** Develop interfaces for other uses of the underlying speech analysis model, such as an accent coach application.
- **OS Application:** Adapting the web-based system into an application that can be run natively on different operation systems (ex: Windows, Linux, macOS).

4 Custom Documents (CDs)

This project is not a research project.

For the fall term, the custom document of choice is the User/Stakeholder/Client interview report. Since the team will be continuously discussing the project scope with our supervisors and stakeholders, we decided this document will most accurately display our progress.

For the Winter term, the custom document of choice is the Usability report. After many discussions with stakeholders, a clear idea of the tests needed to prove the usability of the product will be available.

Appendix — Reflection

1. **What went well while writing this deliverable?**
   Before beginning the deliverable, the group discussed each of the required sections together and took detailed notes for each one. This helped with writing the deliverable because everyone in the group was on the same page and understood the approach we wanted to take. This allowed us to easily divide up the writing work and ensure that each section would be cohesive and unified. Additionally, the group set an internal deadline to complete the writing the day before the official deliverable due date. This helped keep us on track and accountable for our work.

2. **What pain points did you experience during this deliverable, and how did you resolve them?** Since this was the first deliverable and the group's first time working with the LaTeX files in the repo template, there were a few initial challenges and technical difficulties with viewing the PDFs as they would be generated after editing the LaTeX files. We resolved this by communicating our challenges, helping each other, and trying different methods. Additionally, after the initial meeting on the deliverable, there was a little further debate on what features should count as goals vs stretch goals and what the PoC should encompass. This stalled the writing of those respective sections while the group engaged in further discussion to come to a consensus.

3. **How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?** After discussing our project idea with our supervisors, Daniel Pape and Lyn Turkstra (subject matter experts in SLP and speech analysis), we were able to narrow our scope for our PoC model. We came to understand that speech challenges encompasses several different categories and it would likely be difficult to target each one. This led us to narrow the scope to addressing only one subtype of speech challenge (substitutions), with other types as stretch goals. Additionally, we had ideas for multiple types of interfaces that could use our model (an SLP interface, a patient interface, an accent coach interface, etc). We ultimately decided to narrow our scope to focus on the SLP interface, the most important option given our overall project goal: assist speech language pathologists during speech therapy sessions. The other types of interfaces were relegated to stretch goals.

References

[1] Communication Disabilities Access Canada. (2026) People who have communication disabilities. [Online]. Available: https://www.cdacanada.com/resources/communication-disabilities/statistics/

[2] R. Gaines and C. Missiuna, "Early identification: Are toddlers with speech/language impairments at increased risk for developmental coordination disorder?" CanChild Centre for Childhood-Onset Disability Research, McMaster University, 2006. [Online]. Available: https://canchild.ca/resources/96-early-identification-are-toddlers-with-speech-language-impairments-at-increased-risk-for-developmental-coordination-disorder/

[3] American Speech-Language-Hearing Association. (2026) Speech sound disorders: Articulation and phonology. Practice Portal. [Online]. Available: https://www.asha.org/practice-portal/clinical-topics/articulation-and-phonology/

[4] D. P. L. Turkstra, "Initial meeting with supervisors," in-person conversation.